

Interactive collision check path is broken

MC - 295841 Collision

Status	Resolved / Fixed
Affected	1.21.5 Release Candidate 2, 1.21.5, 25w16a, 25w32a
Fixed in	25w32a, 25w37a
Reported	2025-03-25

STEPS TO REPRODUCE

1. Launch Minecraft 1.21.5-rc2.
2. Download and open the attached world.
3. Run `/tp -2.53 76.00 5.53` to move to the setup.
4. Locate the two command blocks at the setup.
5. Activate the command block that spawns a fast TNT entity and highlights the strings it collides with in red wool.
6. Observe the red wool blocks marking the strings triggered by the TNT.
7. Confirm that the interactive collision checks follow a corkscrew path with gaps, rather than continuously following the entity's movement path.

OBSERVED BEHAVIOUR

Actual behavior

Starting with 25w09a, interactive collision checks for an entity that moves along multiple axes in a single tick no longer follow one continuous diagonal path between the entity's starting and ending positions.

Instead, the movement is split into separate cardinal-axis movements, and an interactive collision raycast is performed for each axis independently. As a result, the checked path forms a corkscrew-like shape rather than the direct diagonal path observed in 25w08a and earlier. Because each individual raycast is limited to prevent the crash associated with MC-275437, portions of the axis-aligned segments are omitted, leaving gaps in the interactive-collision path.

Consequently, blocks located along the entity's actual movement path may not be checked. In the attached reproduction, the red wool marks the strings that are successfully triggered by fast-moving TNT; in 25w09a and 1.21.5-rc2, these marks form an incomplete corkscrew with missing sections instead of a continuous path.

This behavior also has the following consequences:

- Interactive collisions for non-projectile entities are checked only along axis-aligned segments, even though the raycast implementation is intended to handle movement across multiple axes.
- The raycast may examine a corner of the corkscrew path outside the entity's direct path. For sufficiently fast diagonal movement, this can cause chunks outside the currently loaded area to be loaded.
- The current approach performs comparatively expensive raycasts for each axis-aligned segment, despite a simple iteration along the relevant coordinate axis being sufficient for those segments.
- Fast-moving entities, including TNT used in TNT-warping contraptions, can therefore trigger interactive blocks inconsistently and cause additional server lag.

EXPECTED BEHAVIOUR

Fast-moving entities should trigger interactive blocks, such as pressure plates, continuously along their actual movement path during a tick, including diagonal movement across all three axes. The collision checks should not contain gaps or deviate from the entity's path, and the block at the movement destination should also be checked, ensuring that all blocks intersected by the movement are activated reliably without causing excessive or unintended chunk loading.

ENVIRONMENT

Minecraft Java Edition Versions: 1.21.5 Release Candidate 2, 1.21.5, 25w16a, 25w32a (includes snapshots)

ORIGINAL DESCRIPTION

When an entity moves, it's supposed to trigger pressure plates and interact with such blocks along its path of motion.

In 24w33a a collision raycast was introduced to create that behavior, however, 25w09a changed it in a bad way.

- Previously (25w08a-) entities moving along all 3 axes in a single tick checked these blocks along their movement path:

(see attached screenshot called 25w08a), in it you can see that the interacted blocks lie all along a diagonal path from start to finish of the single tick movement of an entity. All of the blocks lie along the diagonal path (except the center is cut out, which was done to fix a crash issue MC-275437)

- Currently (25w09a+) entities, while moving along all 3 axes in a single tick, check the blocks along their corkscrew path, like in this picture:

(see attached screenshot called 25w09b), in it you can see that the interacted blocks lie along the corkscrew path of the entity, however the parts of the corkscrew path are missing. This is because now each separate cardinal movement is handled separately and the collision raycast is performed separately for each axis, and as stated above, said raycast is capped to not cause game crashes, thus it skips the middles of the movements along cardinal directions.

How to replicate:

- ■■■■download the attached world, enter it in whichever version you want to observe the behavior (25w08a / 25w09b / 1.21.5-rc2).
- ■■■■run/tp -2.53 76.00 5.53 to get to the setup.
- ■■■■At the setup you'll see 2 command blocks, one will spawn a fast tnt entity and highlight in red all of the strings it collided with in 1 tick. The second command block will clear the volume from red wool, in case you want to trigger the setup a second time, clear the volume before doing it.
- ■■■■Observe the red wool blocks, they indicate which strings were successfully triggered by fast tnt.

Observed behavior:

- interactive collision checks follow a corkscrew path, with gaps in it

Expected behavior:

- Either: fast entities check their path along a diagonal for interactive collisions as they did in 25w08a
- Or: fast entities check their path along a corkscrew for interactive collisions, but without gaps

MANUAL RESULT

Version used	
Reproduced?	YES / NO / PARTIAL
Crash file	
Notes	
Tested by / date	

Live issue: <https://bugs.mojang.com/browse/MC/issues/MC-295841>